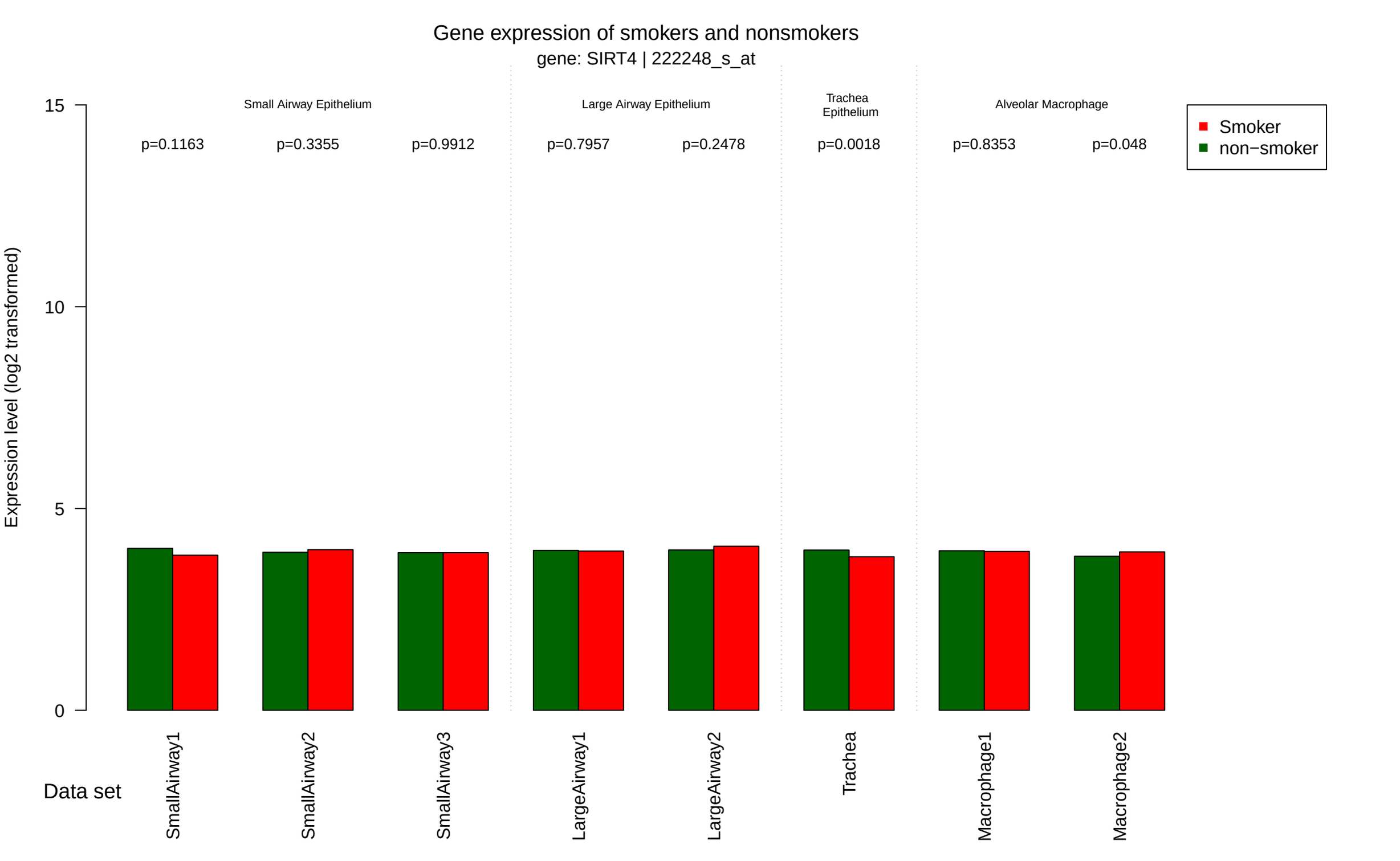
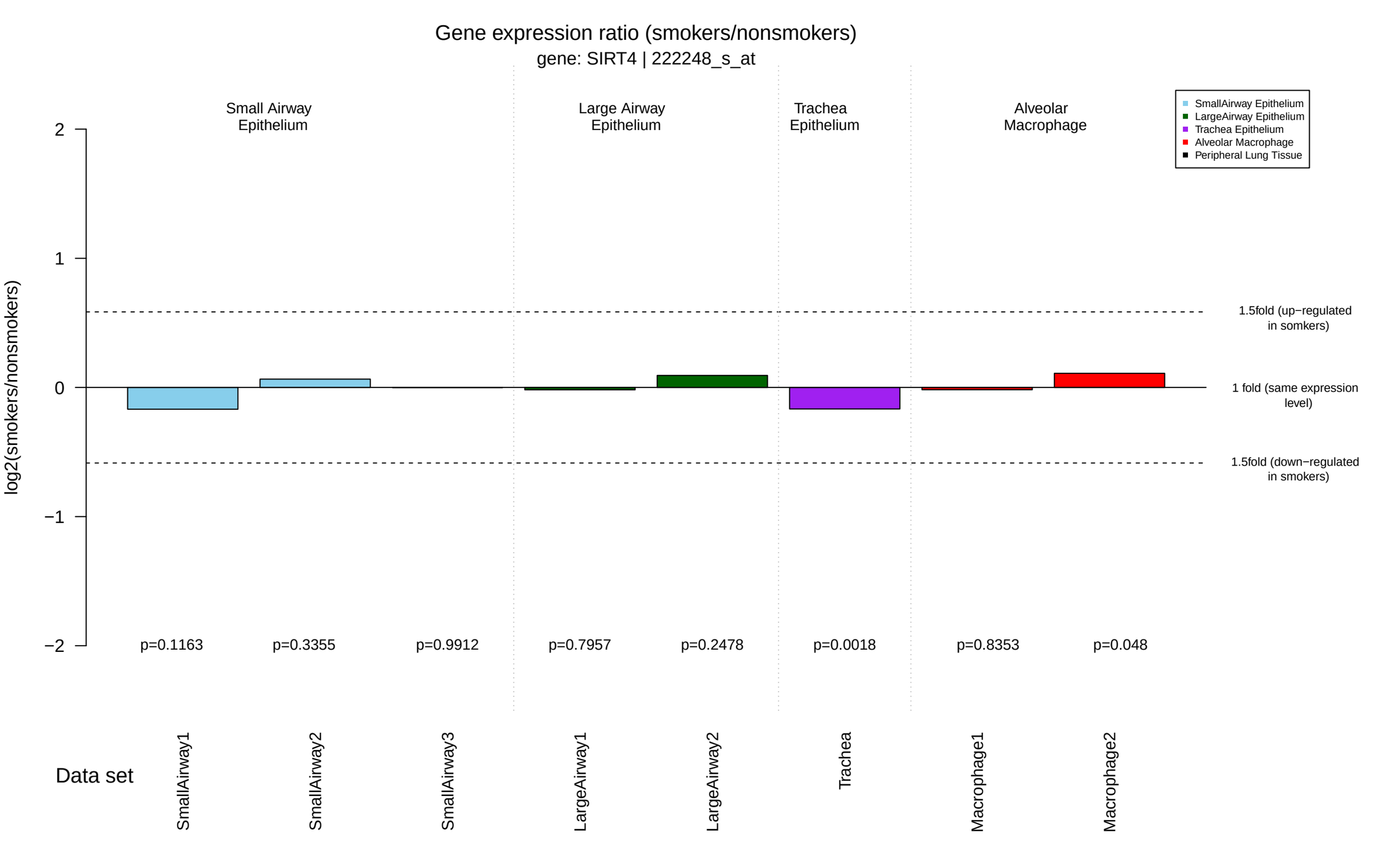
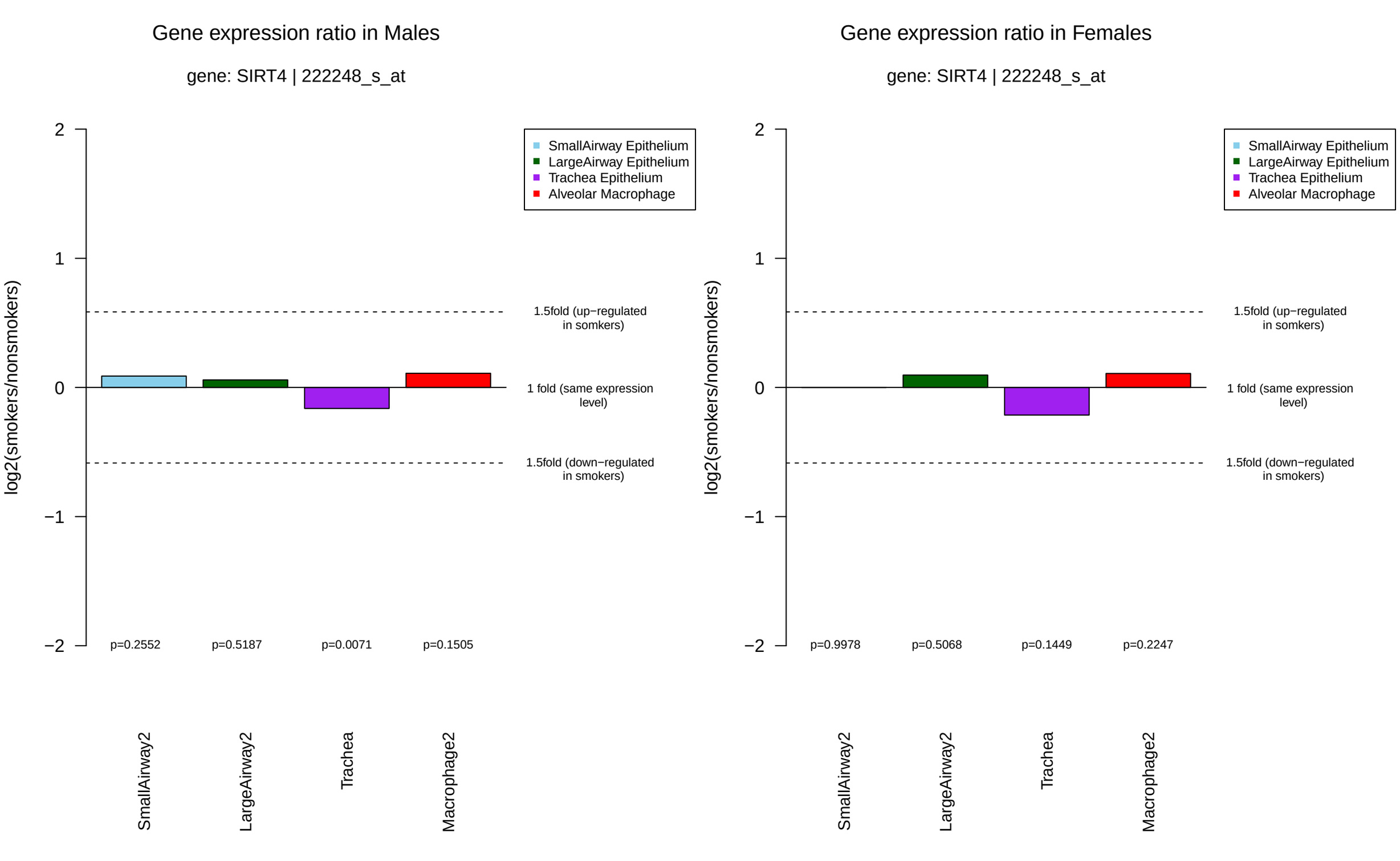
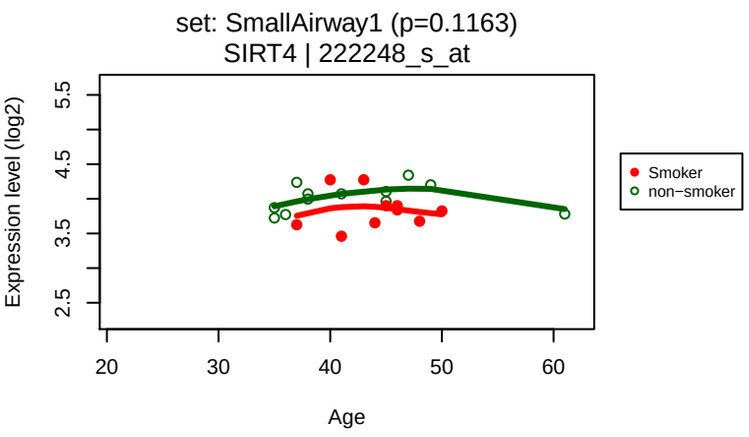
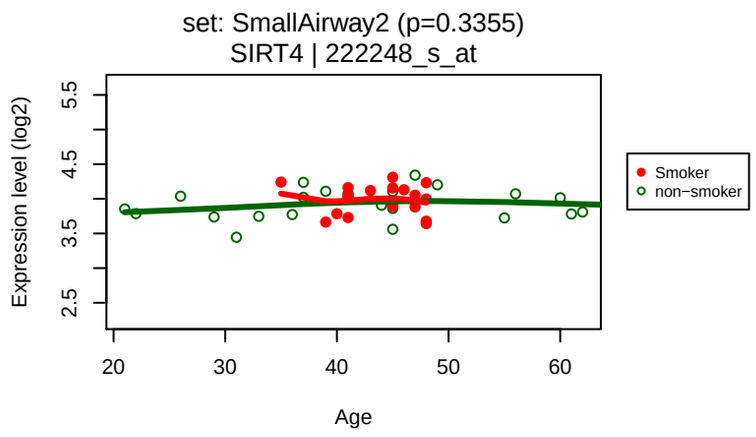
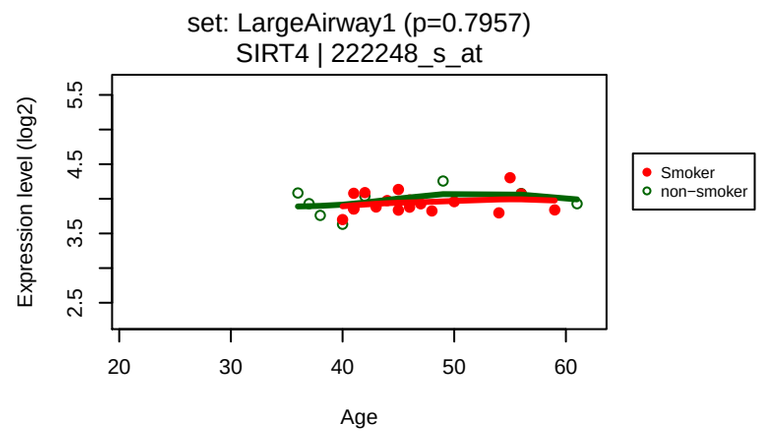

Gene expression of smokers and nonsmokers
gene: SIRT4 | 222248_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.1163
p=0.3355
p=0.9912
p=0.7957
p=0.2478
p=0.0018
p=0.8353
p=0.048
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

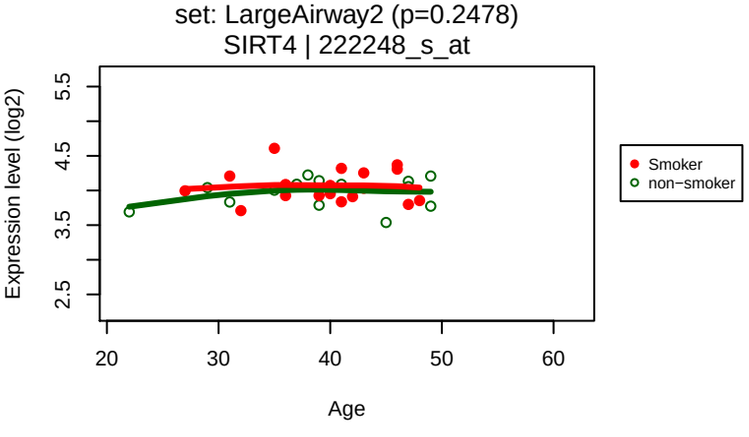

Gene expression ratio (smokers/nonsmokers)
gene: SIRT4 | 222248_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.1163
p=0.3355
p=0.9912
p=0.7957
p=0.2478
p=0.0018
p=0.8353
p=0.048
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

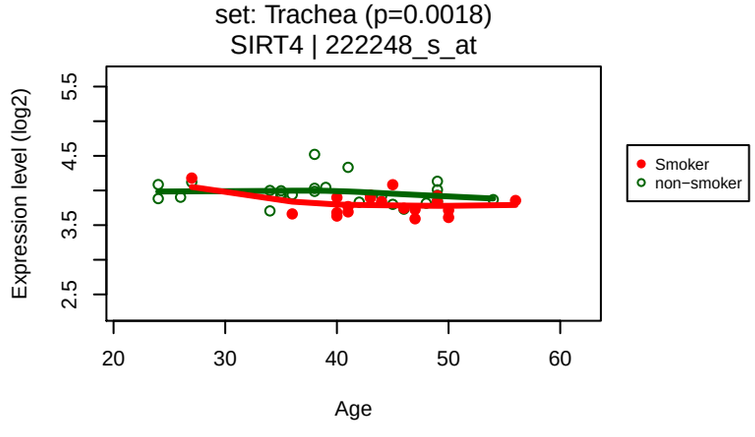

Gene expression ratio in Males
gene: SIRT4 | 222248_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.2552
p=0.5187
p=0.0071
p=0.1505
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: SIRT4 | 222248_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.9978
p=0.5068
p=0.1449
p=0.2247
SmallAirway2
LargeAirway2
Trachea
Macrophage2

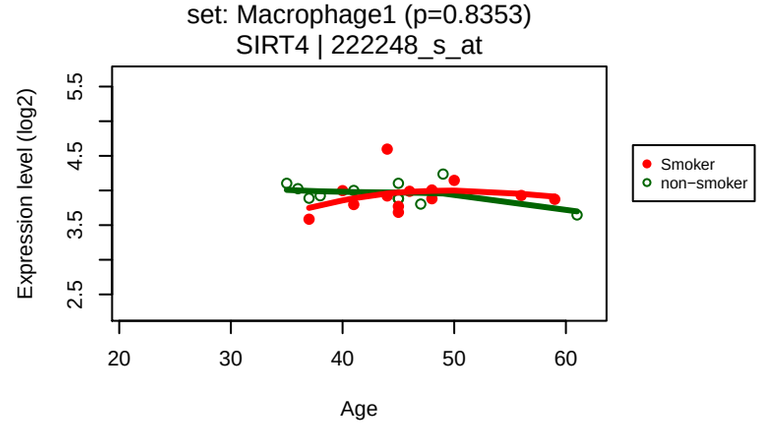

set: SmallAirway1 (p=0.1163)
SIRT4 | 222248_s_at
Expression level (log2)
Age
Smoker
non−smoker

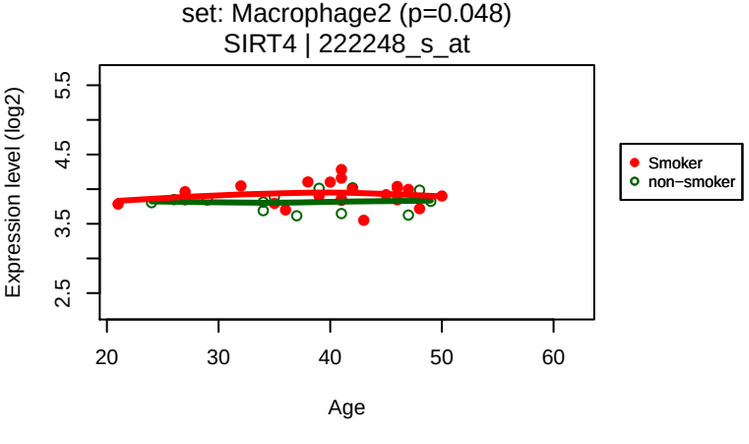

set: SmallAirway2 (p=0.3355)
SIRT4 | 222248_s_at
Expression level (log2)
Age
Smoker
non−smoker

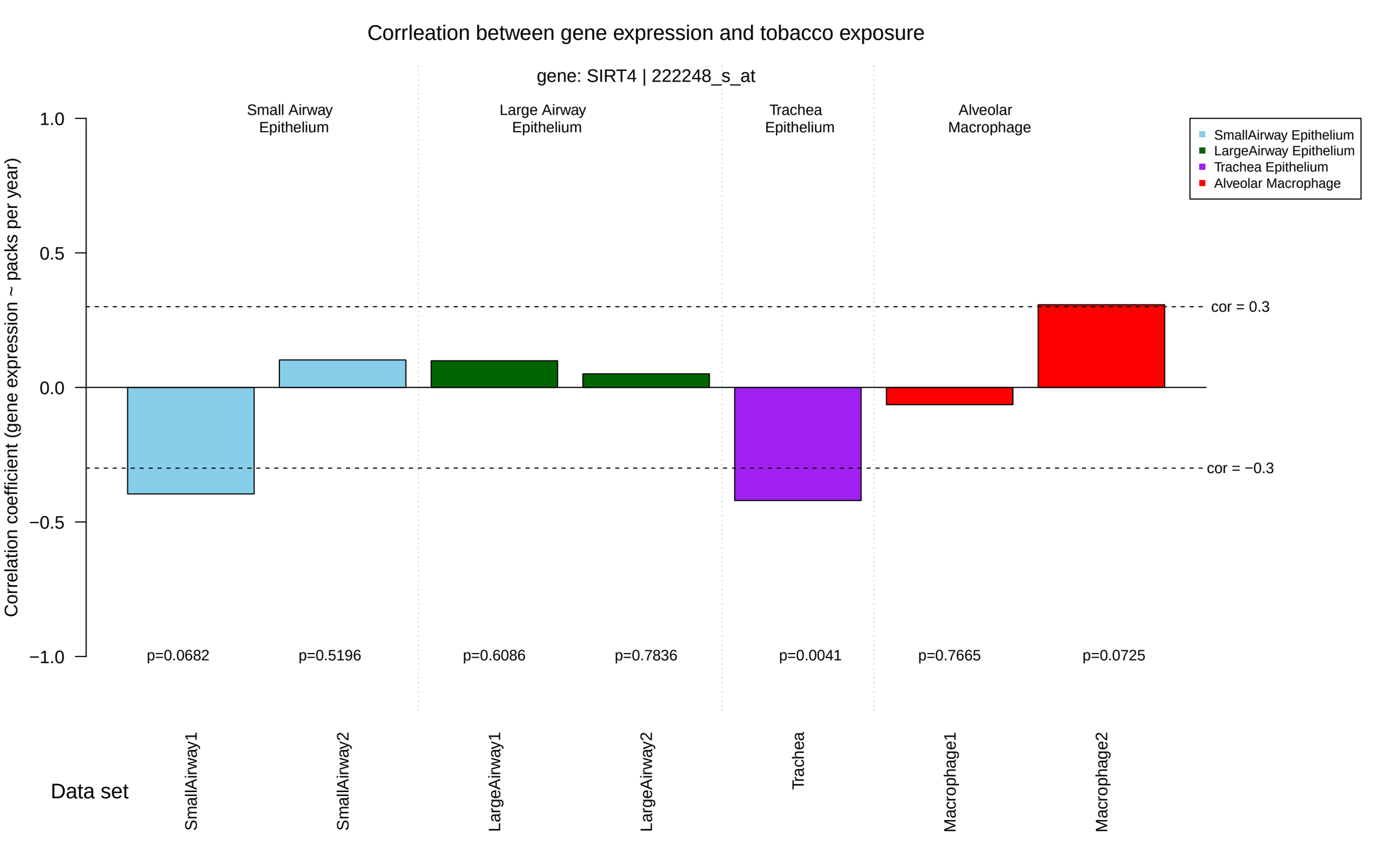

set: LargeAirway1 (p=0.7957)
SIRT4 | 222248_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.2478)
SIRT4 | 222248_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.0018)
SIRT4 | 222248_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.8353)
SIRT4 | 222248_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.048)
SIRT4 | 222248_s_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: SIRT4 | 222248_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.0682
p=0.5196
p=0.6086
p=0.7836
p=0.0041
p=0.7665
p=0.0725
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2